

Solidworks to URDF Export Guide

Ensure the Solidworks-to-URDF Add-In is installed

1. Download and install the sw_urdf_exporter from this link:

https://wiki.ros.org/sw_urdf_exporter

The last ROS 1 release Noetic will go end of life on May 31st with that the ROS Wiki (this website) will also be EOL and transition to being an archive. Maintainers: Please migrate any wiki content into your package's README.md file. If you need more help on migrating code please see this migration guide. Or watch Shane's Lightning Talk from ROSCon 2024.

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sw_urdf_exporter

SolidWorks to URDF Exporter

The SolidWorks to URDF exporter is a SolidWorks add-in that allows for the convenient export of SW Parts and Assemblies into a URDF file. The exporter will create a ROS-like package that contains a directory for meshes, textures and robots (urdf files). For single SolidWorks parts, the part exporter will pull the material properties and create a single link in the URDF. For assemblies, the exporter will build the links and create a tree based on the SW assembly hierarchy. The exporter can automatically determine the proper joint type, joint transforms, and axes.

[Download Installer](#)

If the above provided download fails to work on your system, please install by source before reporting an issue. If installing by source does work, please submit an [Update Installer Request](#)

ROS 2 Documentation

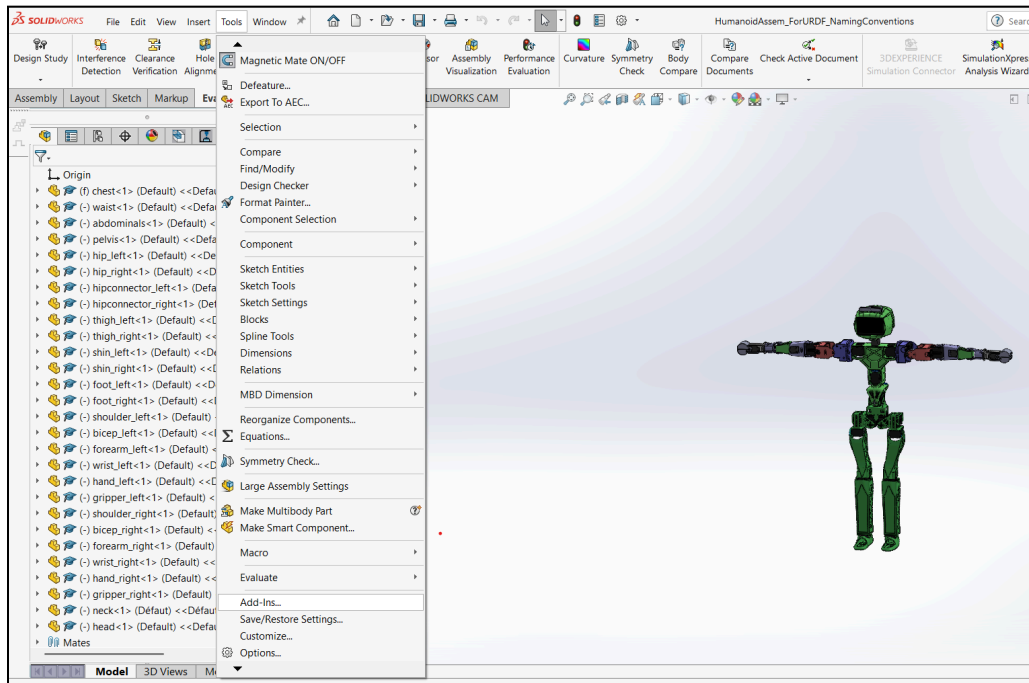
The ROS Wiki is for ROS 1. Are you using ROS 2 (Humble, Iron, or Rolling)?
[Check out the ROS 2 Project Documentation](#)
Package specific documentation can be found on index.ros.org

Wiki

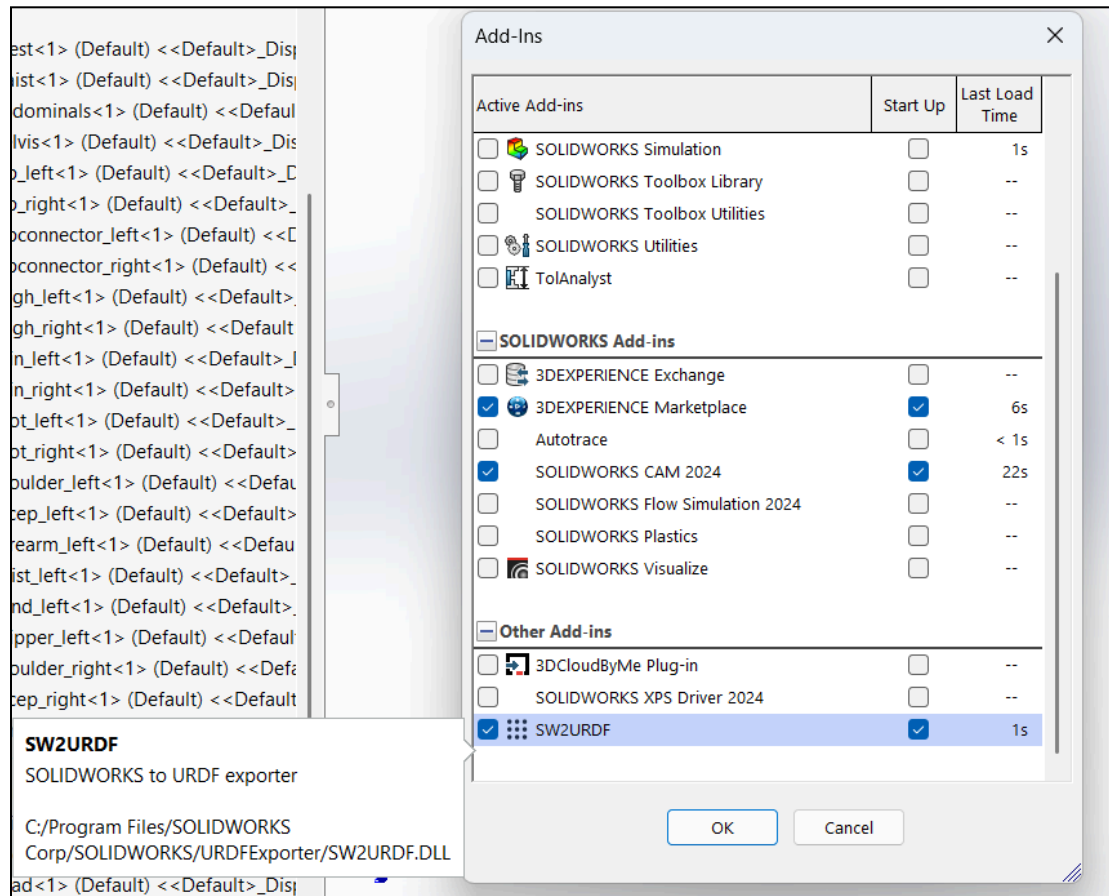
- [Distributions](#)
- [ROS/Installation](#)
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- [sw_urdf_exporter](#)

Page

2. Once you've downloaded and installed, open Solidworks and navigate Tools → Add-Ins



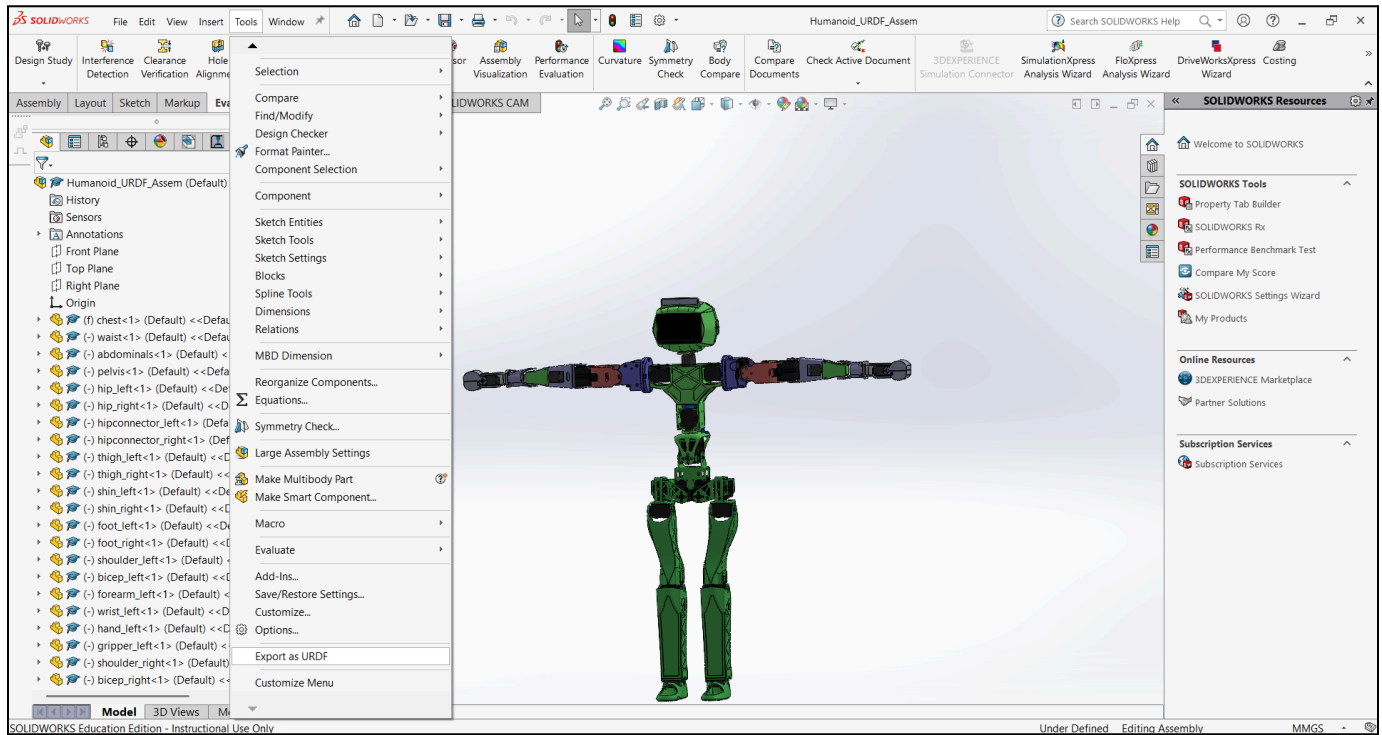
3. Activate the SW2URDF Add-In and click OK



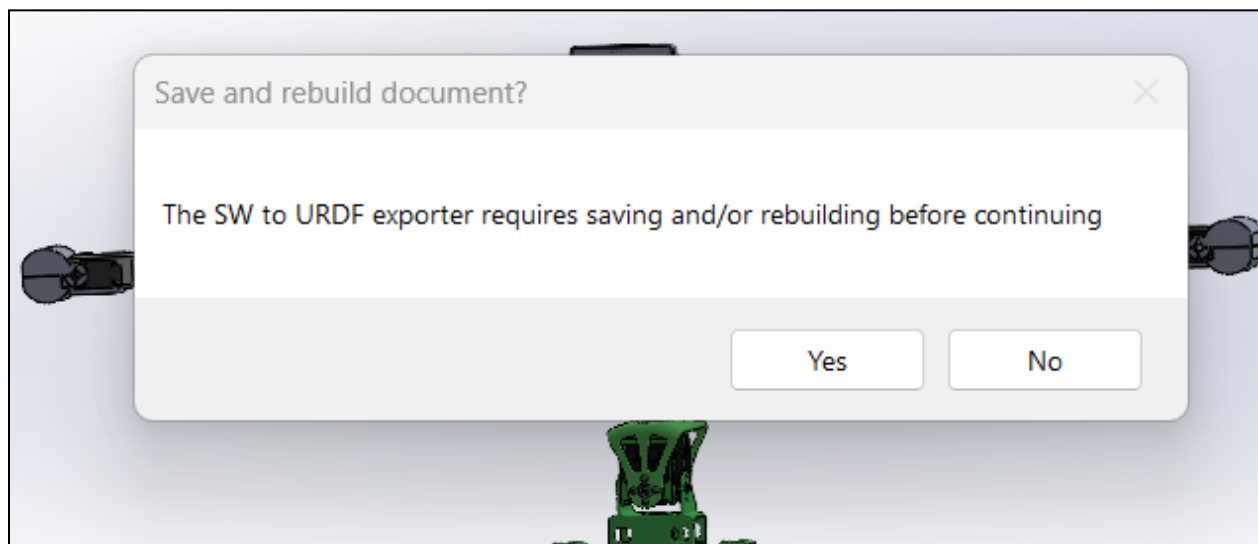
Export Solidworks to URDF

1. Navigate to Tools → Export as URDF

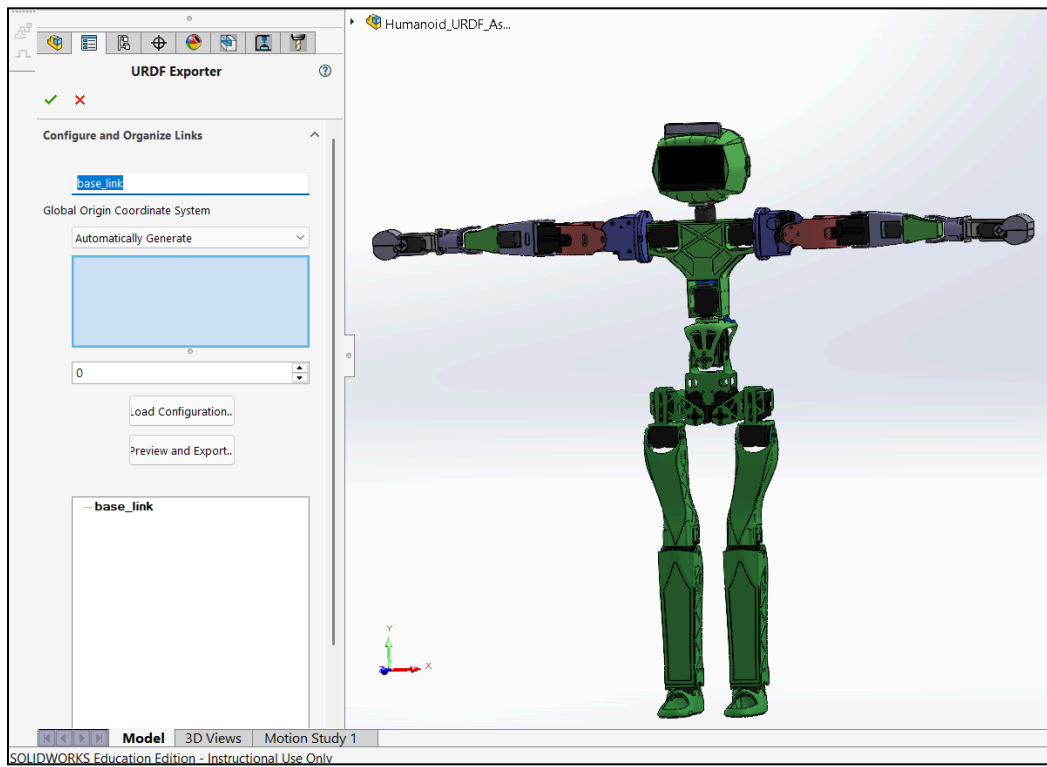
- You may have to hit the down arrow to find this Add-In after it is activated per the last section



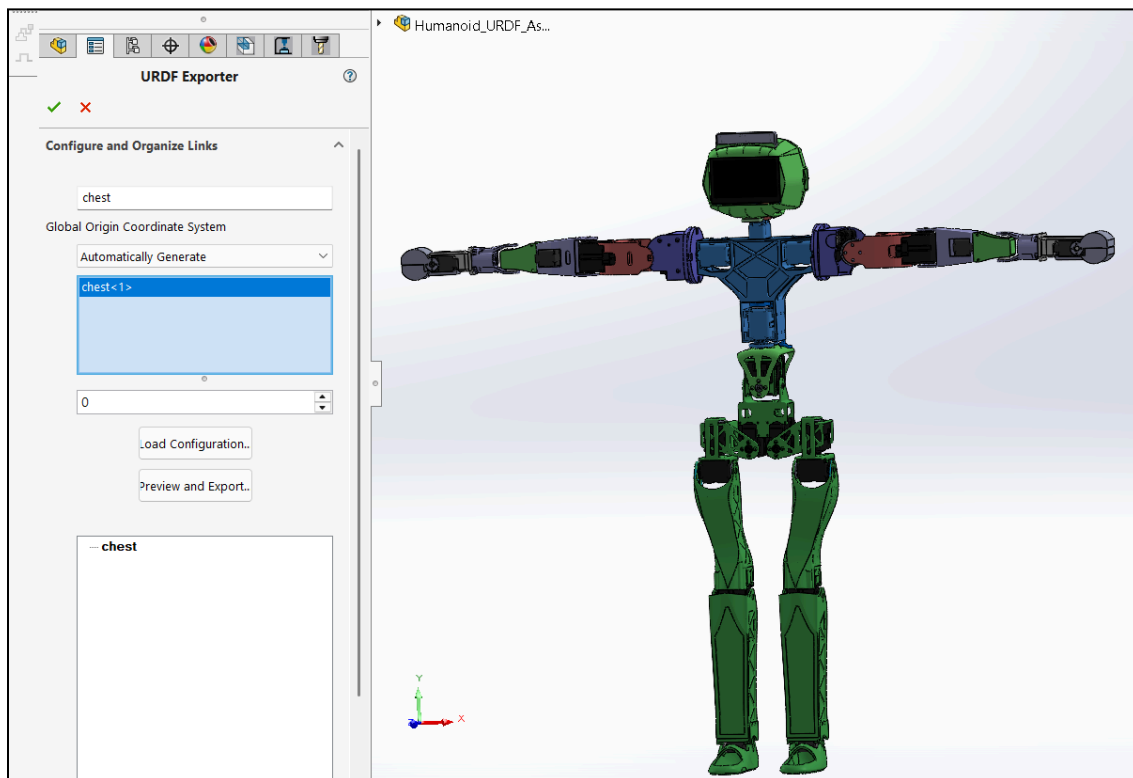
2. Rebuild the document if prompted



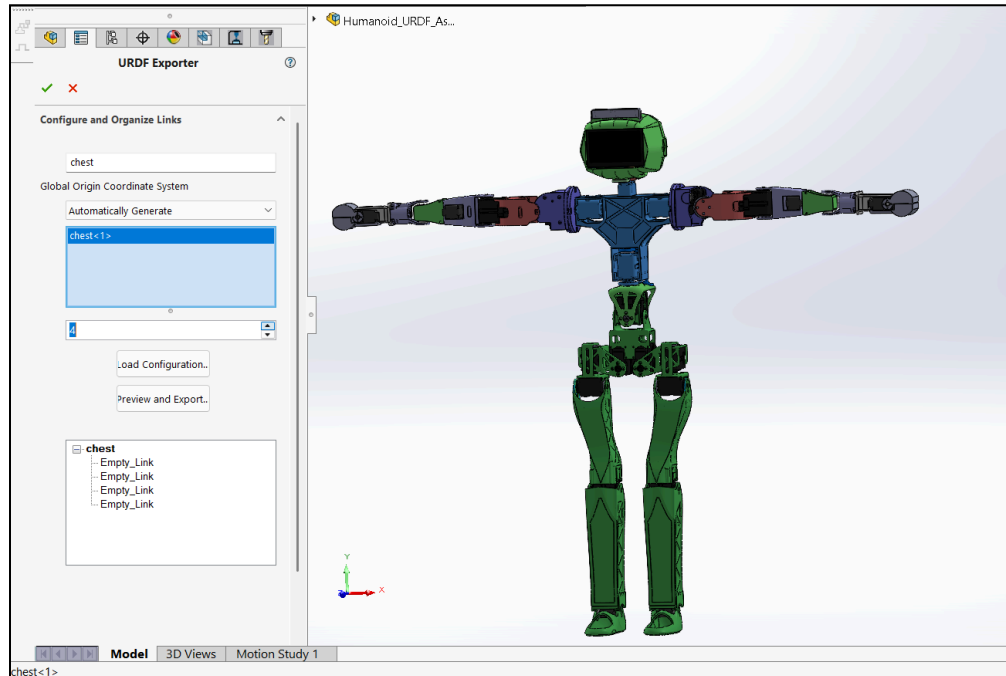
3. Rename “base_link” to “chest” in keeping with naming conventions



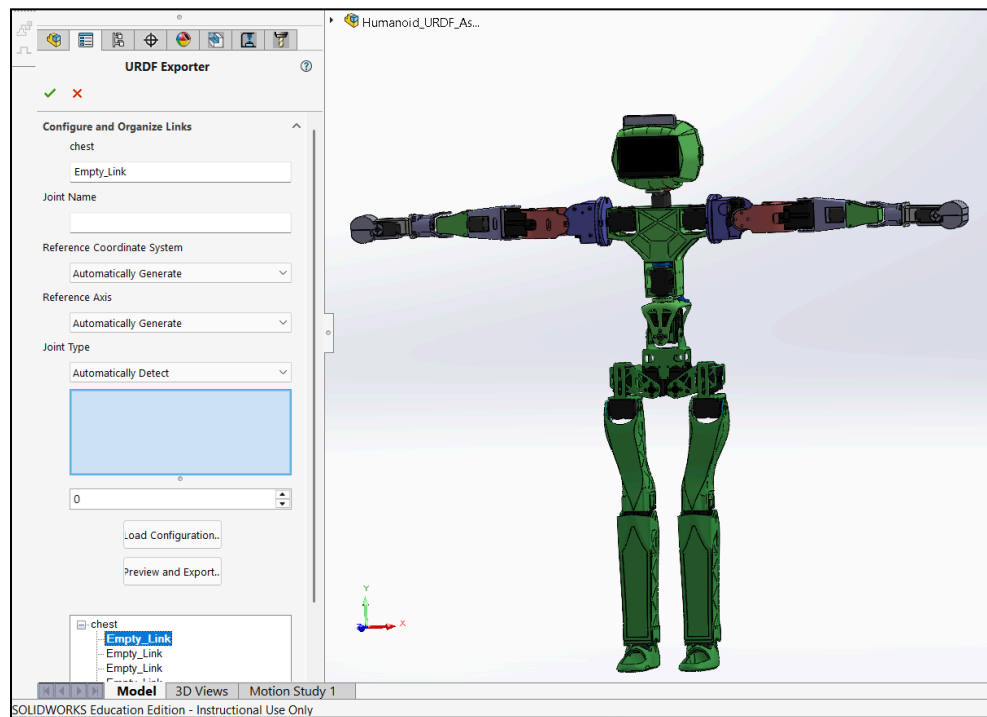
4. Select the “chest” link



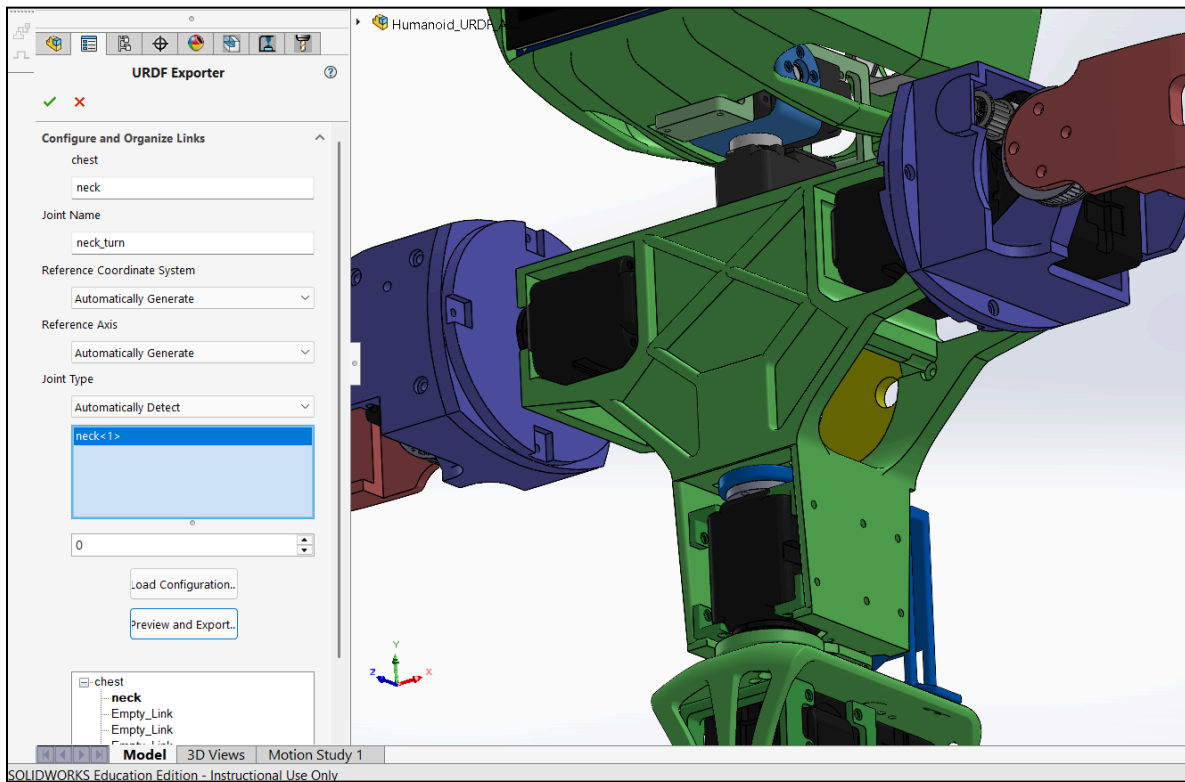
5. Add child links to the URDF tree based on how many joints branch from the base link. For this robot, there are 4 branches off of the chest (base link). These are the head, the right arm, the left arm, and the lower body.



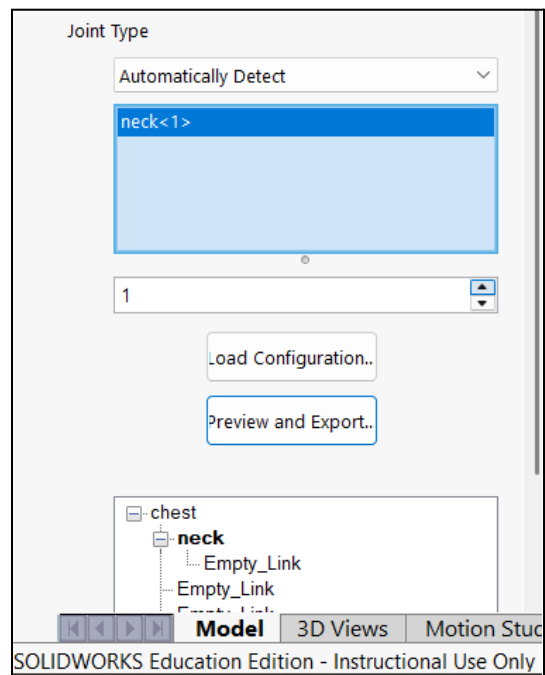
6. Select the first child link



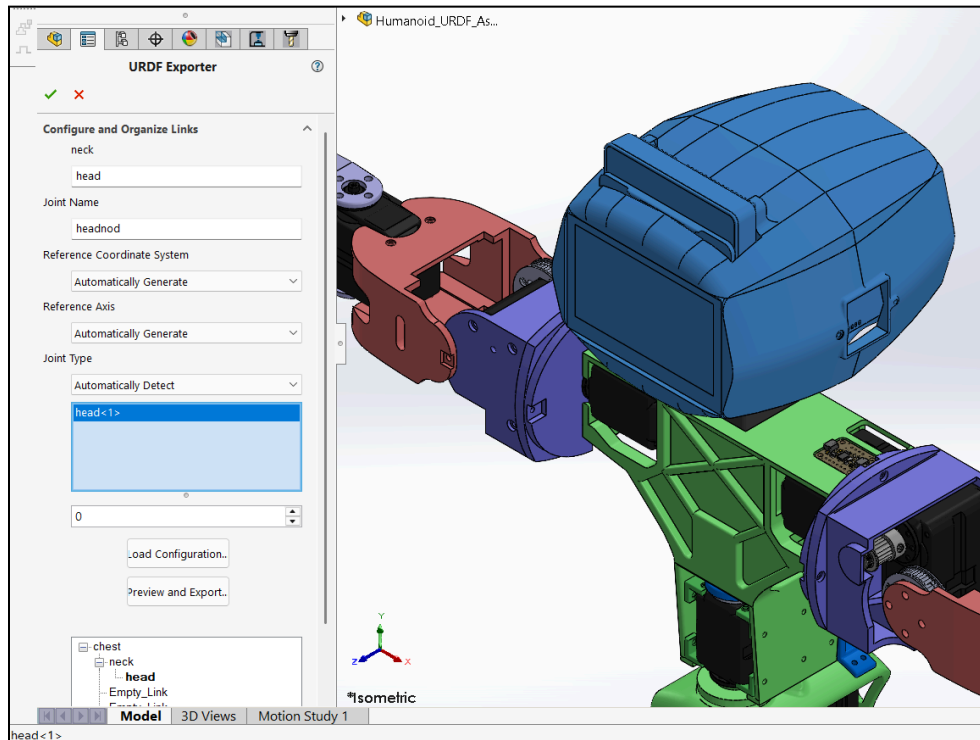
7. Update the link and joint name in accordance with the naming conventions and select the link



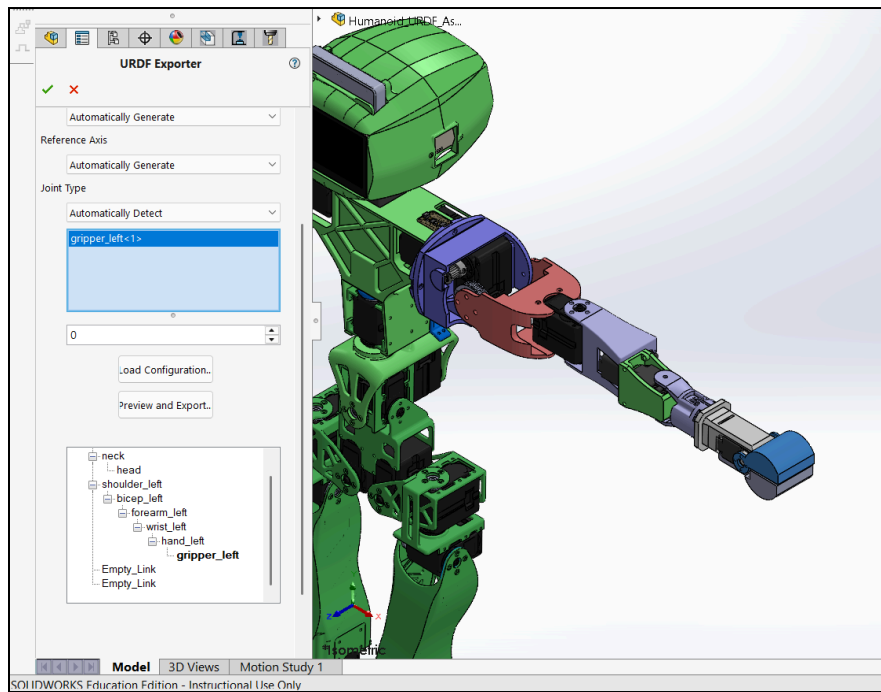
8. Add a child link to the parent link



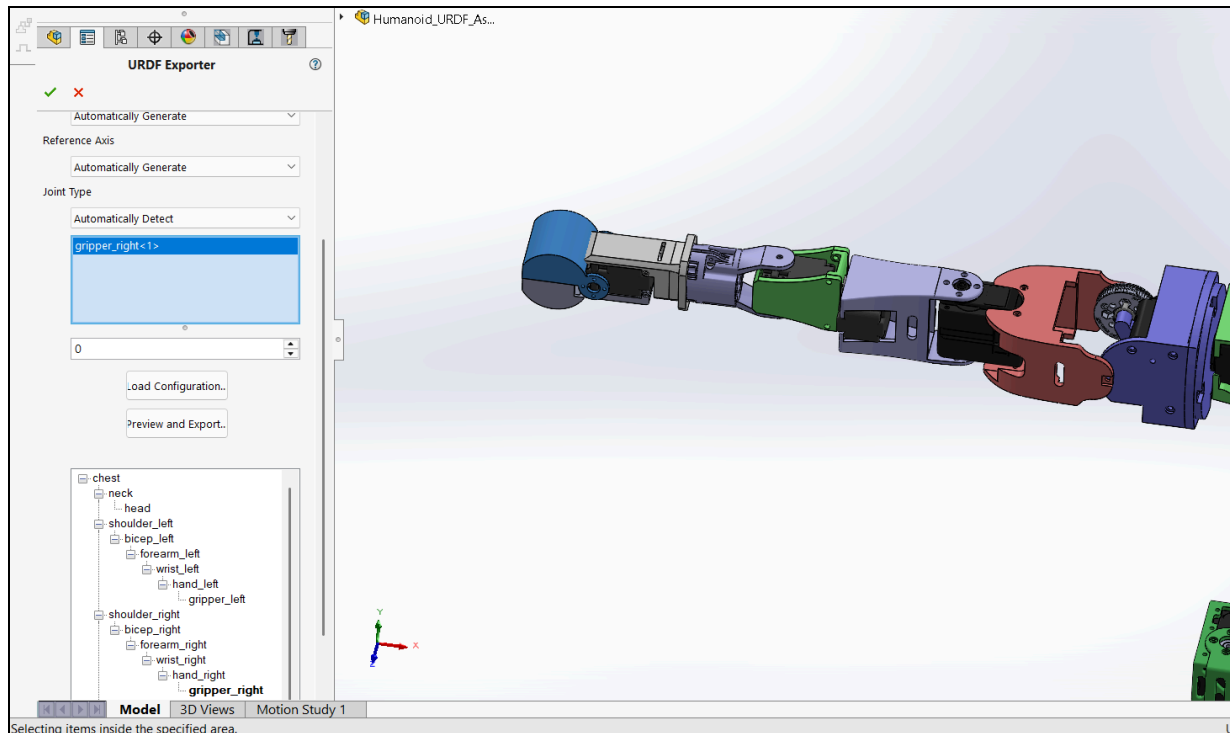
9. Select the child link, define the link name, define the joint name, and select the link model



10. Move on to the arms and the lower body, defining links in sequence according to naming conventions.

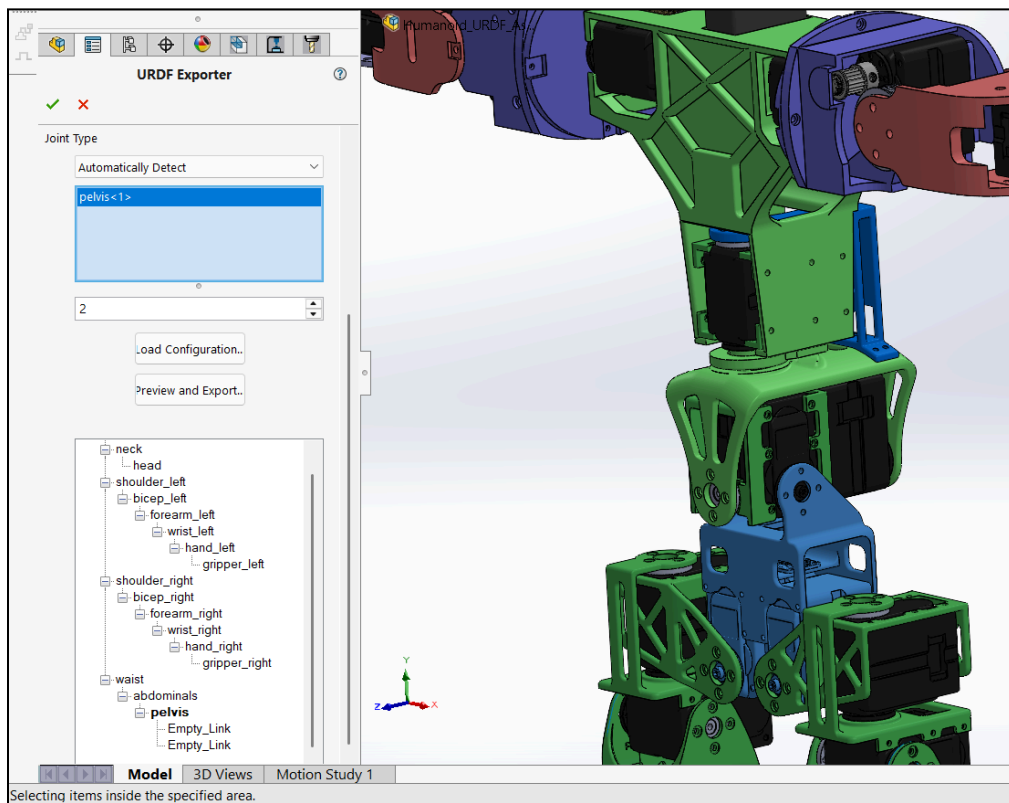


Left arm defined shoulder_left → gripper_left

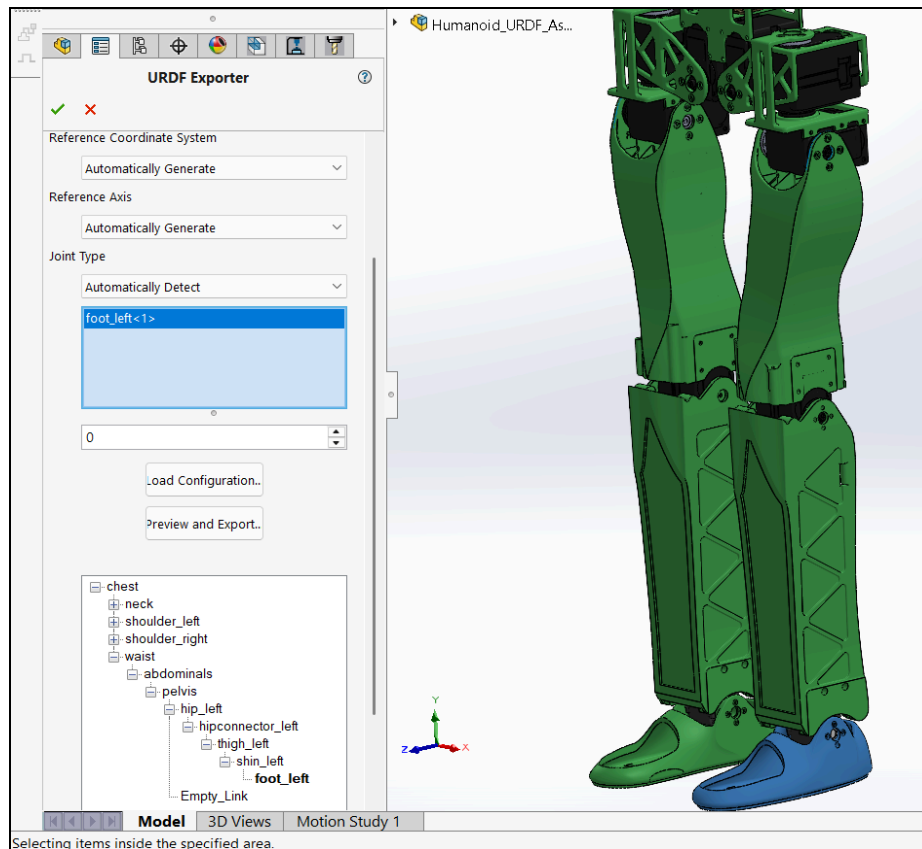


Right arm defined shoulder_right → gripper_right

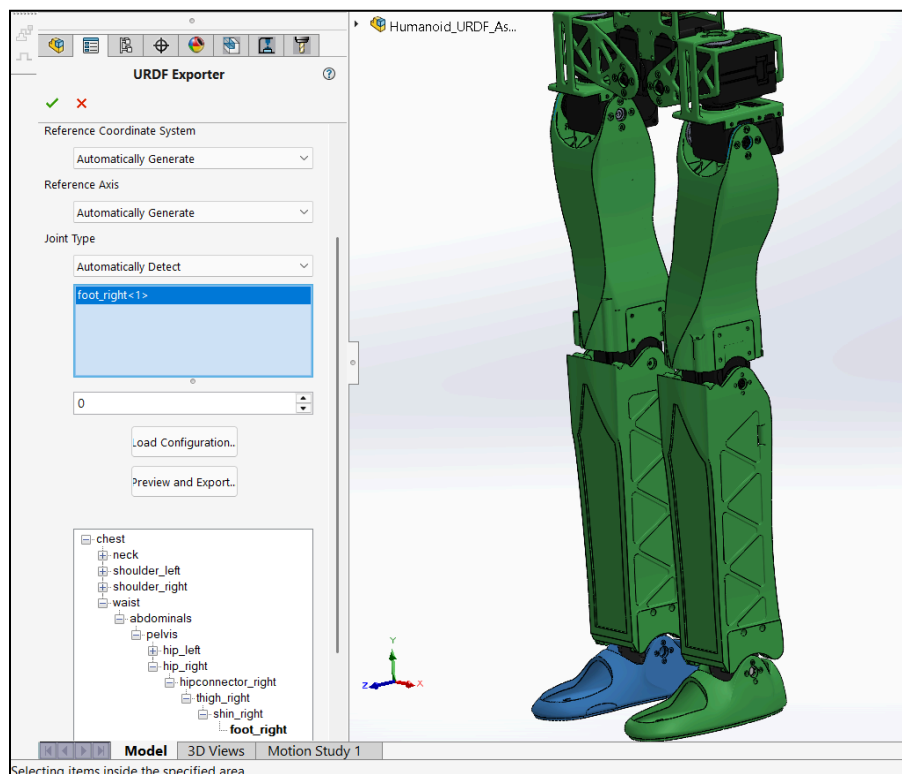
- When you get to a link with multiple child links, add the appropriate number of child links and continue.



Waist, abdominals, and pelvis defined. Pelvis has two child links

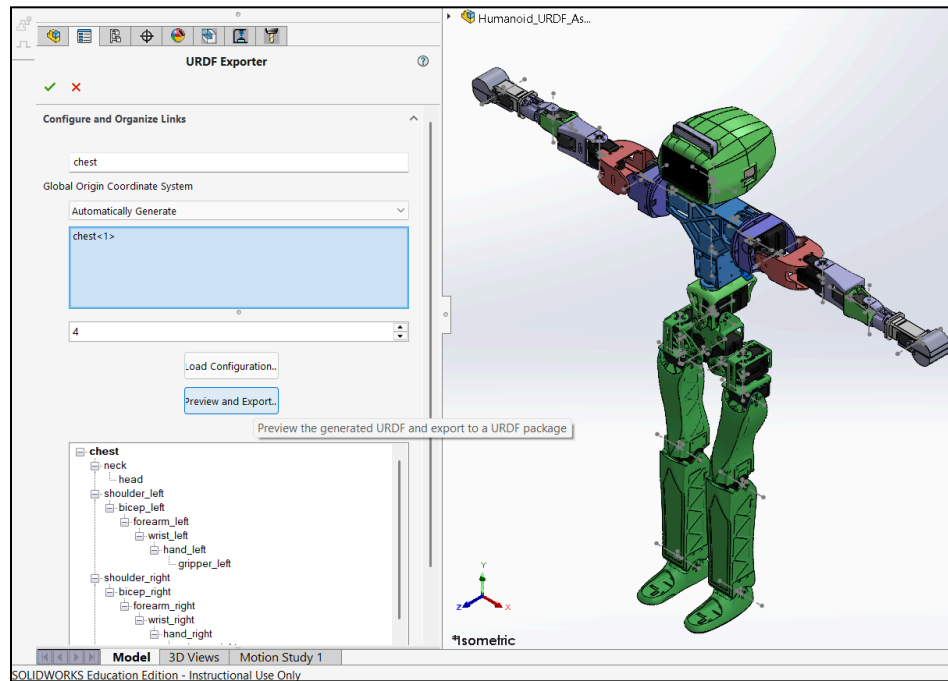


Left leg defined hip_left → foot_left

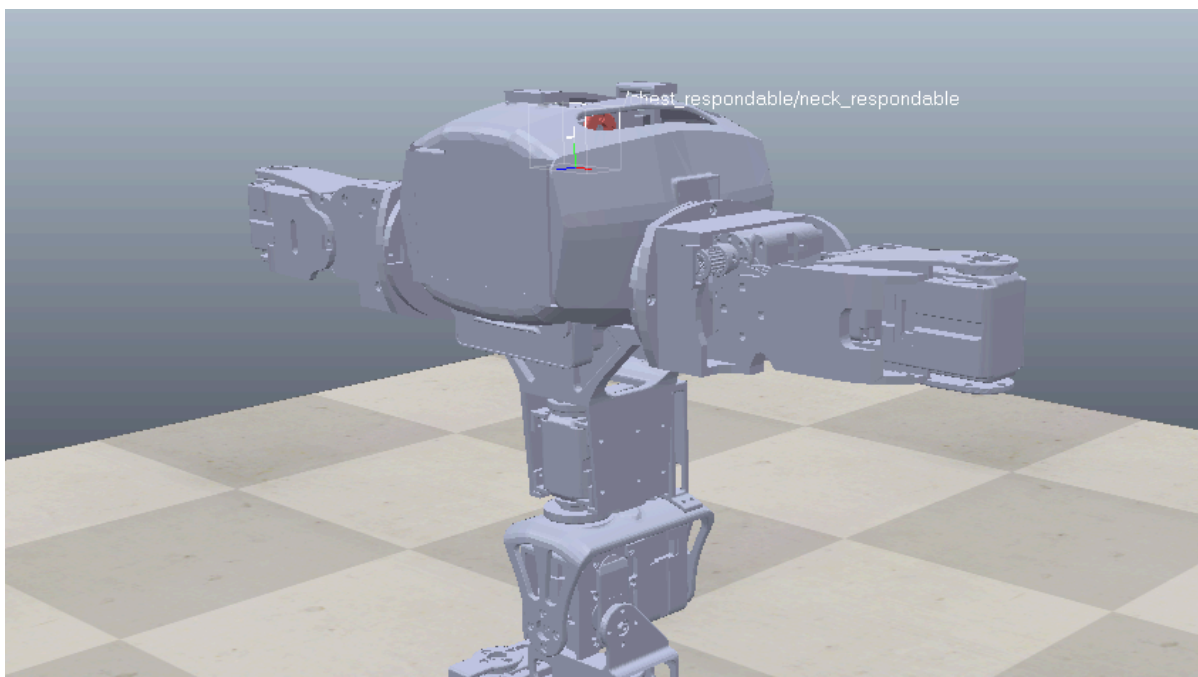


Right leg defined hip_right → foot_right

11. Once the entire chain is defined, click Preview and Export



12. Once the Axes and Origins have generated, fix the headnod, elbow_left, and elbow_right joint orientations. The raw export without these corrections will not have the correct pose. The Roll of headnod must be set to 3.1415, and the Yaw of both elbows must be set to 3.1415.



SolidWorks Assembly to URDF Exporter

Configure Joint Properties

Customize the joint properties. If you want to adjust the coordinate systems and axes in the model, click cancel and restart the export. The tool will recognize your changes on the next run.

neckturn

headnod

shoulderspin_left

biceplift_left

elbow_left

wristspin_left

handcurl_left

gripper_left

shoulderspin_right

biceplift_right

elbow_right

wristspin_right

handcurl_right

gripper_right

chestturn

waistlean

abdominalcrunch

hiplift_left

hiprotate_left

thighlift_left

knee_left

ankle_left

hiplift_right

hiprotate_right

thighlift_right

knee_right

ankle_right

Parent Link: neck

Child Link: head

Joint Name: headnod

Joint Type: continuous

Coordinates: Origin_headnod

Axis: Axis_headnod

Origin*

Position (m)

Orientation (rad)

x

0

Roll

3.1416

y

0.02

Pitch

-1.5708

z

0

Yaw

0

Axis*

x

0

y

0

z

1

Limit

lower (rad)

upper (rad)

effort (N-m)

velocity (rad/s)

Calibration

rising

falling

friction (N-m)

damping (N-m-s/rad)

Dynamics

soft lower limit (rad)

soft upper limit (rad)

k position

k velocity

Safety Controller

Mimic Other Joint

Entries that are blank will not be written to URDF.

* Field group is required

Cancel

Next

SolidWorks Assembly to URDF Exporter

Configure Joint Properties

Customize the joint properties. If you want to adjust the coordinate systems and axes in the model, click cancel and restart the export. The tool will recognize your changes on the next run.

neckturn

headnod

shoulderspin_left

biceplift_left

elbow_left

wristspin_left

handcurl_left

gripper_left

shoulderspin_right

biceplift_right

elbow_right

wristspin_right

handcurl_right

gripper_right

chestturn

waistlean

abdominalcrunch

hiplift_left

hiprotate_left

thighlift_left

knee_left

ankle_left

hiplift_right

hiprotate_right

thighlift_right

knee_right

ankle_right

Parent Link: bicep_left

Child Link: forearm_left

Joint Name: elbow_left

Joint Type: continuous

Coordinates: Origin_elbow_left

Axis: Axis_elbow_left

Origin*

Position (m)

Orientation (rad)

x

0.003175

Roll

0

y

0.1033

Pitch

-1.5708

z

0

Yaw

3.1416

Axis*

x

1

y

0

z

0

Limit

lower (rad)

upper (rad)

effort (N-m)

velocity (rad/s)

Calibration

rising

falling

friction (N-m)

damping (N-m-s/rad)

Dynamics

soft lower limit (rad)

soft upper limit (rad)

k position

k velocity

Safety Controller

Mimic Other Joint

Entries that are blank will not be written to URDF.

* Field group is required

Cancel

Next

SolidWorks Assembly to URDF Exporter

Configure Joint Properties

Customize the joint properties. If you want to adjust the coordinate systems and axes in the model, click cancel and restart the export. The tool will recognize your changes on the next run.

- neckturn
 - headnod
- shoulderspin_left
 - biceplift_left
 - elbow_left
 - wristspin_left
 - handcurf_left
 - gripper_left
 - shoulderspin_right
 - biceplift_right
 - elbow_right**
 - wristspin_right
 - handcurf_right
 - gripper_right
 - chestturn
 - waistlean
 - abdominalcrunch
 - hiplift_left
 - hiprotate_left
 - thighlift_left
 - kneelift_left
 - ankle_left
 - hiplift_right
 - hiprotate_right
 - thighlift_right
 - kneelift_right
 - ankle_right

Parent Link: bicep_right
Child Link: forearm_right

Joint Name: Joint Type:

Coordinates:

Axis:

Origin*		Orientation (rad)		Axis *		Limit	
	Position (m)						
x	0.003175	Roll	0	x	1	lower (rad)	
y	0.1033	Pitch	-1.5708	y	0	upper (rad)	
z	0	Yaw	3.14159	z	0	effort (N-m)	
						velocity (rad/s)	

Calibration

rising

falling

Dynamics

friction (N-m)

damping (N-m-s/rad)

Safety Controller

soft lower limit (rad)

soft upper limit (rad)

k position

k velocity

☐ Mimic Other Joint

Entries that are blank will not be written to URDF.
*Field group is required

13. Once the axes have been fixed, click Next, then Export URDF and Meshes and export to a desired file location. This will create a folder with subfolders containing meshes, the .urdf file, and other useful files for simulations.

SolidWorks Assembly to URDF Exporter

Configure Link Properties

Use this page to make any changes to the links' properties

- chest
 - neck
 - head
 - shoulder_left
 - bicep_left
 - forearm_left
 - wrist_left
 - hand_left
 - gripper_left
 - shoulder_right
 - bicep_right
 - forearm_right
 - wrist_right
 - hand_right
 - gripper_right
 - waist
 - abdominals
 - pelvis
 - hip_left
 - hipconnector_left
 - thigh_left
 - shin_left
 - foot_left
 - hip_right
 - hipconnector_right
 - thigh_right
 - shin_right
 - foot_right

Inertial

Origin (m)

Mass (kg)

Moment of Inertia (Kg * m²)

x Roll ixx ixy ixz

y Pitch iyy iyz

z Yaw izz

Visual and Collision Meshes

Origin (m)

Color

Red

Green

Blue

Alpha

Mesh Detail

☒ Course ☐ Fine

Material name

Texture (Replaces Color)

14. Finally, open the .urdf file in a text editor and correct the mesh file path as follows. This will ensure the meshes are accessible when collaborators pull the exported files from github.

```
<visual>
  <origin
    xyz="0 0 0"
    rpy="0 0 0" />
  <geometry>
    <mesh
      filename="package://Humanoid_URDF_9-10/meshes/chest.STL" />
    </geometry>
  <material
    name="">
    <color
      rgba="0.792156862745098 0.819607843137255 0.933333333333333 1" />
    </material>
</visual>
```

Incorrect file path

```
<visual>
  <origin
    xyz="0 0 0"
    rpy="0 0 0" />
  <geometry>
    <mesh
      filename="../meshes/chest.STL" />
    </geometry>
  <material
    name="">
    <color
      rgba="0.792156862745098 0.819607843137255 0.933333333333333 1" />
    </material>
</visual>
```

Corrected universal file path

15. Upload to desired URDF compatible simulator!